ChatGPT

# Probability of Range Selection

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Source: https://chatgpt.com/c/698c1351-0064-832b-b18f-f49ad060105c

If a population has a mean of 125 with a standard deviation of 25, what is the probability of randomly selecting a value in the range of 75-175?

Assuming the population is **normally distributed** with mean $\mu = 125$ and standard deviation $\sigma = 25$:

Compute z-scores:

$$z_1 = \frac{75 - 125}{25} = \frac{-50}{25} = -2, \qquad z_2 = \frac{175 - 125}{25} = \frac{50}{25} = 2.$$

So the probability is

$$P(75 \le X \le 175) = P(-2 \le Z \le 2) = \Phi(2) - \Phi(-2) = 0.9772498681 - 0.0227501319 = 0.954$$

Answer: **≈ 0.9545**, or **95.45%** (about 95.4%).

(If you don't assume normality, Chebyshev's theorem only guarantees at least $1 - 1/2^2 = 75\%$ will lie within 2 standard deviations.)

Two different normal distributions could have the same mean but different standard deviation.

Question 4 options:
True
False

**Answer: True**

Two normal distributions can have the **same mean** but **different standard deviations**.

- The **mean** decides the center of the distribution.
- The **standard deviation** decides how spread out the data is.

So both curves can be centered at the same point, but one can be wider (larger standard deviation) and one narrower (smaller standard deviation).

Therefore, the correct answer is **True**.